Practice workbook · Product rule and chain rule: differentiating products and 'function of a function'

Using the chain rule on composite functions and the product rule on products such as an exponential times a polynomial, a trig function or a root.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-CAL-01

1. Question 6 · part a.i · 3 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part a.i only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x - 3)^2}$ with respect to x (3)

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1. Question 6 · part a.i · 3 marks · Working and answer

Working and answer

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2. Question 1 · part whole · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part whole only.

1 Differentiate with respect to x

$$6e^{3x^2}\cos 2x$$

(3)

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(Total for Question 1 is 3 marks)

2. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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3. Question 11 · part a · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

11 Given that

$$y = \frac{e^{4x}}{32}(8x^2 - 4x + 1)$$

(a) show that $\frac{dy}{dx} = x^2e^{4x}$

(5)

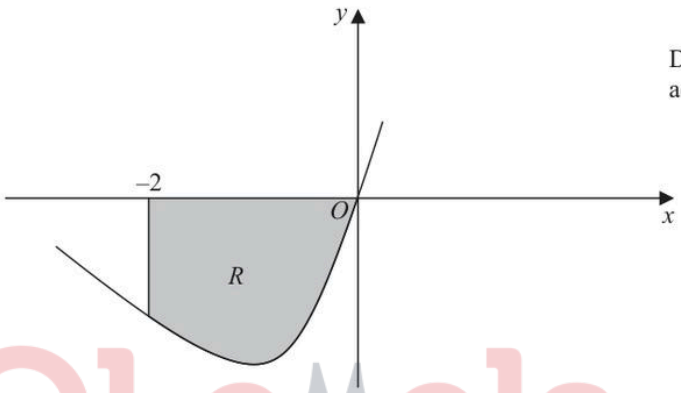


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows part of the curve C with equation $y = 3xe^{2x}$

The finite region R bounded by C, the straight line with equation $x = -2$ and the x-axis, shown shaded in Figure 2, is rotated through 360° about the x-axis.

(b) Using part (a), find the volume, to 2 significant figures, of the solid formed.

(4)

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3. Question 11 · part a · 5 marks · Working and answer

Working and answer

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4. Question 3 · part a · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

3 Differentiate with respect to x

(a) $e^{2x}\sqrt{5x-3}$

(3)

(b) $\dfrac{x^3}{\cos 3x}$

(3)

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4. Question 3 · part a · 3 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

6 (a) Given that $y = (4x - 3)e^{2x}$

(i) find $\frac{dy}{dx}$ (3)

(ii) show that $(4x - 3)\frac{dy}{dx} = (8x - 2)y$ (2)

(b) Differentiate $\frac{\sin 5x}{(x-3)^2}$ with respect to x (3)

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 8 marks)

5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 7 · part a · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

7 $y = e^{2x} \cos 2x$

(a) Show that

$$\frac{dy}{dx} = 2y - 2e^{2x} \sin 2x$$

(4)

(b) Hence show that

$$\frac{d^2y}{dx^2} = 4\frac{dy}{dx} - 8y$$

(5)

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6. Question 7 · part a · 4 marks · Working and answer

Working and answer

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7. Question 9 · part a · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

9

$$y = e^{-4t} \cos 2t$$

(a) Show that $2e^{-4t}\sin 2t = -\frac{dy}{dt} - 4y$ (3)

Given that $\frac{d^2y}{dt^2} + M\frac{dy}{dt} + Ny = 0$ where M and N are integers

(b) find the value of M and the value of N (5)

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7. Question 9 · part a · 3 marks · Working and answer

Working and answer

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8. Question 3 · part a · 3 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

3 Given that $y = e^{2x}(x^2 + 1)$

(a) find $\frac{dy}{dx}$

(3)

The straight line l is the tangent to the curve with equation $y = e^{2x}(x^2 + 1)$ at the point on the curve where $x = 0$

(b) Find an equation for l in the form $y = mx + c$

(3)

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8. Question 3 · part a · 3 marks · Working and answer

Working and answer

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9. Question 5 · part a · 5 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

5 Differentiate with respect to x

(a) $e^{4x}(6x+2)^{\frac{3}{2}}$

Give your answer in the form $e^{4x}\left(\sqrt{6x+2}\right)(Ax+B)$ where A and B are integers.

(5)

(b) $\dfrac{\sin 3x}{(2x-4)^3}$

(3)

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9. Question 5 · part a · 5 marks · Working and answer

Working and answer

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10. Question 9 · part a · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 Given that $f(x) = (1+x^2)^4$

(a) show that $f'(x) = 8x(1+x^2)^3$ (2)

The curve C has equation $y = \dfrac{\sin 3x}{(1+x^2)^4}$

The point A on C has x coordinate $\dfrac{2\pi}{3}$

(b) Show that the gradient of the normal to C at A is

$$-\frac{1}{3}\left(1+\left(\frac{2\pi}{3}\right)^2\right)^4$$ (7)

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10. Question 9 · part a · 2 marks · Working and answer

Working and answer

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